

Statistical chronometry of meteorites revisited

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Abstract

The chronology of the early solar system rests on extinct ($^{26}\text{Al}/^{26}\text{Mg}$, $^{53}\text{Mn}/^{53}\text{Cr}$, $^{182}\text{Hf}/^{182}\text{W}$) and extant ($^{207}\text{Pb}/^{206}\text{Pb}$) chronometers, which should agree for rapidly cooled meteorites from isotopically homogeneous parts of the solar system. Testing this concordance requires a statistical framework. Here we introduce two approaches to statistical chronometry of meteorites. In a first approach, we revisit the weighted-mean based approach of Desch et al. (2023, doi:10.1016/j.icarus.2023.115607 and doi:10.1016/j.icarus.2023.115611), which tracks events forward in time from the start of the solar system, t_{SS} . This model was originally applied to a compilation of 38 multi-dated meteorites whose Pb–Pb dates were adjusted for $^{238}\text{U}/^{235}\text{U}$ fractionation. After fixing an error in the model’s original formulation, we show that the adjusted database exhibits statistically significant overdispersion with respect to the analytical uncertainties. A second approach is to build the statistical chronology counting backward in time, starting from the present and converting extinct-nuclide dates to ‘Pb–Pb time’ through a single offset parameter estimated by weighted regression with a fixed unit slope. Applying the second algorithm to the 38-sample data compilation yields the same conclusion of statistically significant overdispersion. Extending the regression to a random-effects model allows the excess dispersion to be quantified. It is equivalent to up to 7.5% heterogeneity in initial $^{26}\text{Al}/^{27}\text{Al}$, $^{53}\text{Mn}/^{55}\text{Mn}$ or $^{182}\text{Hf}/^{180}\text{Hf}$, or up to 380 kyr of excess scatter in Pb–Pb dates. Whether the cause of the excess dispersion is astrophysical or analytical remains unresolved. Routine measurement of $^{238}\text{U}/^{235}\text{U}$ in Pb–Pb dated mineral phases and more realistic statistical models for isochron selection may help to separate these causes.

1 Introduction

The chronology of the early solar system can be constrained from the radiogenic decay products of extinct (^{26}Al , ^{53}Mn , ^{182}Hf) and extant (^{235}U , ^{238}U) nuclides in meteorites. The extinct nuclides form the basis of relative chronometers, which can be used to estimate the age difference between pairs of meteorites. In principle, the extant nuclides can be used to anchor meteorites in absolute time, counting back from the present. However, in practice the precision of such absolute dates is limited by the analytical uncertainty of the decay constants (Ludwig, 2000). For the $^{207}\text{Pb}/^{206}\text{Pb}$ method, this uncertainty amounts to ca. 9 Myr (2σ) at 4567 Ma (Figure 1). The decay constant uncertainty is greatly reduced when calculating the difference between two similar $^{207}\text{Pb}/^{206}\text{Pb}$ dates (Figure 1). For example, a 10 Myr age difference near the onset of the solar system is associated with a decay constant uncertainty of only 7 kyr. Therefore, when comparing $^{207}\text{Pb}/^{206}\text{Pb}$ dates within the 4550–4568 Ma age range, decay constant uncertainties can be safely ignored, turning the $^{207}\text{Pb}/^{206}\text{Pb}$ -method into a relative chronometer with similar precision as the extinct chronometers.

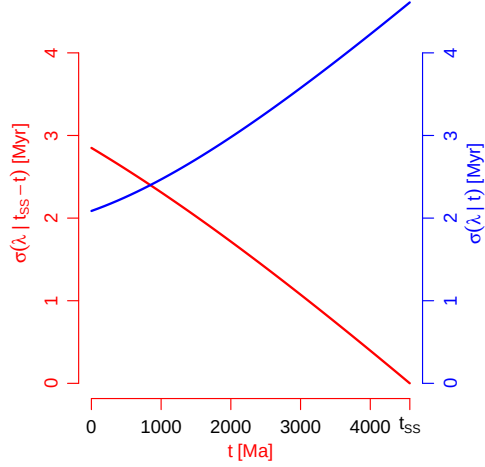


Figure 1: Decay constant uncertainty as a function of the $^{207}\text{Pb}/^{206}\text{Pb}$ age t (blue); and the difference between a meteorite of age t and the start of the solar system t_{SS} (red).

For rapidly cooled meteorites in isotopically homogenised regions of the early solar system, extinct and extant chronometers should be concordant. Jointly considering a basket of such double-dated meteorites leads to the establishment of a ‘statistical chronology’ of those meteorites. This paper presents two approaches to this problem. A first approach is to establish a chronology for the early solar system counting forward in time from the start of the solar system, t_{SS} , using the scatter of double-, triple- and quadruple-dated meteorites around their weighted means (Section 2). A second approach is to build the statistical chronology counting backward in time, starting from the present and using linear regression instead of weighted means (Section 3).

Section 4 applies both approaches to a collection of 11 rapidly cooled meteorites compiled by Desch et al. (2023a,b). Previous statistical analysis of this database using the first approach suggested concordance of the Al–Mg, Mn–Cr, Hf–W and Pb–Pb methods. After fixing an error in Desch et al. (2023a,b)’s calculations, we come to a different conclusion and find a small but statistically significant amount of excess dispersion in the data. Section 5 quantifies this overdispersion by modifying the linear regression method.

This paper will use the following definitions:

t_{SS} : the start of the solar system, measured from the present;

t_{Al} , t_{Pb} , t_{Mn} , t_{Hf} : the Al–Mg, Pb–Pb, Mn–Cr and Hf–Cr date, respectively, measured from the present;

$\Delta t_* = t_{SS} - t_*$, where $*$ $\in \{Al, Pb, Mn, Hf\}$;

λ_{26} , λ_{53} : the decay constant of ^{26}Al and ^{53}Mn , respectively;

$[^y\text{X}/^x\text{X}]_{SS}$: the $^y\text{X}/^x\text{X}$ (e.g., $^{27}\text{Al}/^{26}\text{Al}$)-ratio of the solar system at t_{SS} ;

$[^y\text{X}/^x\text{X}]_0$: the $^y\text{X}/^x\text{X}$ -ratio of a meteorite at the time of its formation;

$\sigma[*]$: the analytical uncertainty (estimated standard deviation) of $*$;

χ^2 : the chi-square statistic for homogeneity;

ν : the number of degrees of freedom of the chi-square distribution under the null hypothesis;

MSWD: the mean square weighted deviation, defined as $\text{MSWD} = \chi^2/\nu$.

2 Approach 1: $t_{SS} \rightarrow t$

Desch et al. (2023a) define t_{SS} as the time when the $^{26}\text{Al}/^{27}\text{Al}$ -ratio of the Solar System was $(^{26}\text{Al}/^{27}\text{Al})_{SS} = 5.23 \times 10^{-5}$, assuming isotopic homogeneity. The actual value of t_{SS} is then estimated by minimising the sum of the squared differences between $\Delta t_{Al} = \frac{1}{\lambda_{26}} \left(\ln \left[\frac{^{26}\text{Al}}{^{27}\text{Al}} \right]_{SS} - \ln \left[\frac{^{26}\text{Al}}{^{27}\text{Al}} \right]_0 \right)$ and $\Delta t_{Pb} = t_{SS} - t_{Pb}$ for a ‘basket’ of double-dated meteorites that are thought to have achieved simultaneous isotopic closure for both systems. Given a set of n double-dated meteorites, Desch et al. (2023a) test the concordance of the Al–Mg and Pb–Pb dates by finding the value of t_{SS} that minimises a chi-square statistic:

$$\chi^2 = \sum_{i=1}^n \sum_{* \in \{Al, Pb\}} \left(\frac{\Delta t_{*,i} - \Delta \bar{t}_i}{\sigma[t_{*,i}]} \right)^2 \quad (1)$$

where $\bar{t}_i$ is the error-weighted mean of the Al–Mg and Pb–Pb dates of the i^{th} meteorite. If all n meteorites are concordant, and if analytical uncertainty is the only source of dispersion, then χ^2 is expected to follow a chi-square distribution with ν degrees of freedom. This expectation can be used to formally test the concordance of the data: low χ^2 -values indicate homogeneity, high values ($\chi^2 \gg \nu$) indicate overdispersion. Desch et al. (2023a) posit that $\nu = 2n - 1$, assuming that $\Delta t_{Al,i}$ and $\Delta t_{Pb,i}$ are statistically independent. This is incorrect. In reality, $\Delta t_{Al,i}$ and $\Delta t_{Pb,i}$ are perfectly anti-correlated, so that the correct number of degrees of freedom is $\nu = n - 1$. This value accounts for the fact that χ^2 comprises n weighted means of paired values (each of which represents one degree of freedom) minus one degree of freedom because t_{SS} is estimated from the data.

Equation 1 can be generalised to more than two chronometers. Suppose that the collection of n meteorites comprises of n_2 double-dated, n_3 triple-dated and n_4 quadruple-dated meteorites, then Desch et al. (2023b) posit that $\nu = 2n_2 + 3n_3 + 4n_4 - m$, where m is the number of nuisance parameters to be estimated as part of the optimisation among t_{SS} , $[^{53}\text{Mn}/^{55}\text{Mn}]_{SS}$, $[^{182}\text{Hf}/^{180}\text{Hf}]_{SS}$ and λ_{53} . This is also incorrect, with the correct value being $\nu = n_2 + 2n_3 + 3n_4 - m$. This value accounts for the fact that the weighted mean of N values has $N - 1$ degrees of freedom, so that the sum of the χ^2 -values for n_N weighted means has $(N - 1)n_N$ degrees of freedom.

3 Approach 2: $t \leftarrow \text{present}$

Approach 1 defines t_{SS} in terms of an assumed value for $(^{26}\text{Al}/^{27}\text{Al})_{SS}$, and using a basket of double-dated meteorites. Future updates to the basket would result in new estimates for t_{SS} . However, this is not the only option. A mathematically equivalent use of the basket would be to fix t_{SS} and treat $(^{26}\text{Al}/^{27}\text{Al})_{SS}$ as an unknown. In fact, the choice of t_{SS} is completely arbitrary and can be avoided altogether by expressing the Al/Mg-dates in ‘Pb/Pb-time’, by counting time from the present instead of from t_{SS} . This can be done by finding the offset A that minimises the misfit between t_{Al} and t_{Pb} , where

$$t_{Al} = A + \frac{1}{\lambda_{26}} \ln \left(\frac{^{26}\text{Al}}{^{27}\text{Al}} \right)_0 \quad (2)$$

in which $A = t_{SS} - \ln(^{26}\text{Al}/^{27}\text{Al})_{SS} / \lambda_{26}$. Any change to the basket of double-dated meteorites would change A and, hence, t_{Al} . However, t_{Pb} would remain unaffected, unlike Desch et al. (2023a)’s Δt_{Pb} and Δt_{Al} , which would both change.

The offset parameter A can be estimated by weighted least squares regression with a fixed unit slope. The dispersion of the data around the 1:1 line is quantified by a chi-square statistic with $n - 1$ degrees of freedom, i.e. exactly as under Approach 1. Mn–Cr and Hf–W data can be treated in the same way.

4 Application to the Desch et al. (2023b) database

Desch et al. (2023a,b) applied Approach 1 to the compilation of meteorites shown in Table 1. In their detailed survey of the literature, they adjusted the published results of several studies in different three ways. First, they recalculated the Pb–Pb dates of some meteorites by retrospectively adjusting the $^{238}\text{U}/^{235}\text{U}$ -ratios by 0.3‰ to account for possible U-fractionation between pyroxene and whole rock (Tissot et al., 2017). Second, they recalculated the Pb–Pb isochrons for certain meteorites using a different selection of aliquots than reported in the original publications. Third, they treated the ^{53}Mn -decay constant as an unknown and optimised the λ_{53} -value that brought the Mn–Cr data as close as possible to concordance with the other chronometers.

Applying Equation 1 to the corrected data of Table 1, Desch et al. (2023b) report a χ^2 -value of 47.94 for the 38 meteorite dates, assuming $\nu = 37 - 4 = 34$ degrees of freedom. Under this assumption, the p-value for the chi-square test is 0.057, which is greater than 0.05 and fails to reject the null hypothesis that the meteorite ages are all concordant. However, as discussed in Section 2, the correct number of degrees of freedom is not 34 but $\nu = 6 \times 1 + 6 \times 2 + 2 \times 3 - 4 = 20$. Using this correct value yields a p-value of 0.00043, resulting in a rejection of the null hypothesis.

Desch et al. (2023b) also present a second χ^2 -value of 35.97 with $\nu = 33$ degrees of freedom by removing the Mn–Cr date of NWA 4801. This results in a p-value of 0.33, which they deem compatible with the basket of meteorites achieving concordant isotopic closure for all four isotopic systems. However, combining the same statistic with the correct number of degrees of freedom ($\nu = 19$) produces a p-value of 0.011, which contradicts this conclusion.

Table 1: Summary table for the database of Desch et al. (2023b). Headers use the following shorthand notation: $Al_0 \equiv (^{26}\text{Al}/^{27}\text{Al})_0$, $Mn_0 \equiv (^{53}\text{Mn}/^{55}\text{Mn})_0$, $Hf_0 \equiv (^{182}\text{Hf}/^{180}\text{Hf})_0$; t_{orig} stands for the original Pb–Pb dates extracted from the literature; and t_{adj} represents the adjusted values used by Desch et al. (2023b).

sample	Al_0	$2\sigma(Al_0)$	Mn_0	$2\sigma(Mn_0)$	Hf_0	$2\sigma(Hf_0)$	t_{orig}	$2\sigma(t_{orig})$	t_{adj}	$2\sigma(t_{adj})$
D’Orbigny	3.93	0.39	3.23	0.03	7.15	0.17	4564.42	0.12	4563.24	0.21
SAH 99555	3.64	0.18	3.28	0.17	6.87	0.15	4564.614	0.131	4563.51	0.24
NWA 1670	5.92	0.59	2.85	0.92			4565.39	0.24	4564.02	0.66
NWA 1296					7.01	0.28	4564.2	0.45	4563.02	0.45
LEW 86010			1.35	0.05	4.8	0.42	4558.55	0.15		
NWA 4590			0.85	0.4	4.63	0.17	4557.76	0.38	4557.57	0.38
NWA 4801			0.96	0.04	4.52	0.16	4556.91	0.21	4556.63	0.28
Angra do Reis			1.1	0.4	4.02	0.24	4556.45	0.29	4556.26	0.29
NWA 2999					5.43	0.34	4560.74	0.47	4560.55	0.47
Asuka 881394	13.07	0.56	3.86	0.23			4564.95	0.53	4564.76	0.53
Ibitira			1.06	0.5			4556.75	0.57	4556.56	0.57
NWA 7325	3.03	0.14					4563.9	1.7	4563.7	1.7
NWA 2976	4.05	0.15					4563.16	0.57	4563.16	0.57
NWA 6704	3.15	0.38	2.59	0.34			4562.76	0.26	4562.76	0.26

We now proceed to subject Table 1 to Approach 2. Figure 2 compares the original Pb–Pb dates with the Al–Mg, Mn–Cr and Hf–W dates, respectively. For each of the three pairs of measurements, the offset A was found by minimising the weighted sum of the square residuals around the 1:1 line, using the algorithm of Vermeesch (2024). All three cases exhibit high MSWD-values and fail the chi-square test for homogeneity. This indicates that the data are overdispersed with respect to the analytical uncertainties.

The excess dispersion of the data around the 1:1 line in Figure 2 can be attributed to either (i) $(^{26}\text{Al}/^{27}\text{Al})_{ss}$ -heterogeneity (Krestianinov et al., 2023); (ii) partial resetting of the $^{207}\text{Pb}/^{206}\text{Pb}$ -clock during transient heating events (Desch et al., 2023a); (iii) $^{238}\text{U}/^{235}\text{U}$ -fractionation (Tissot et al., 2017); or (iv) inappropriate outlier rejection prior to regression of the Al–Mg, Mn–Cr, Hf–W or Pb–Pb data (Desch et al., 2023a; Vermeesch, 2026).

Desch et al. (2023a,b) adjusted the Pb–Pb dates in Table 1 to remove factors (ii) and (iii). This adjustment does indeed lower the MSWD-value considerably (Figure 3). Nevertheless, the three pairwise comparisons still fail the chi-square test. This either means that the dispersion is caused by scenario (i), or that the adjustment has not gone far enough. In any case, Approach 2 leads to the same conclusion as Approach 1, namely that the different chronometers are not concordant.

Section 5 will extend Approach 2 to treat the various sources of heterogeneity as an unknown, whose effect can be estimated from the data.

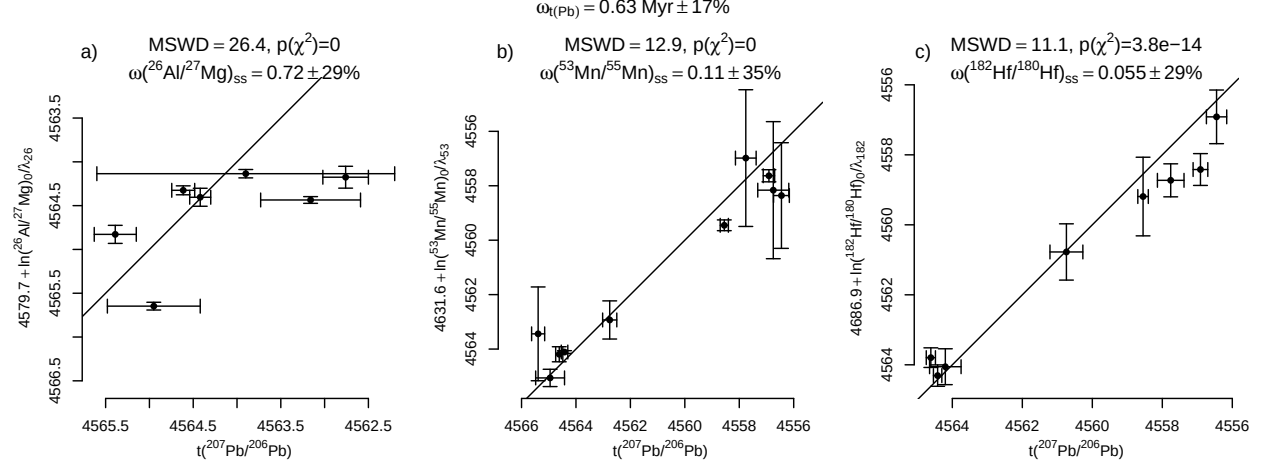


Figure 2: Pairwise comparison of the original Pb–Pb dates of Table 1 with Al–Mg (panel a), Mn–Cr (panel b) and Hf–W (panel c) dates for the same, rapidly cooled meteorites. The 1:1 lines were obtained using the optimal A -values (Equation 2). All three datasets are overdispersed around this line with respect to the analytical uncertainties ($\text{MSWD} \gg 1$). ω -values are the estimated standard deviations of the excess dispersion, assuming normality (Section 5). This dispersion can either be attributed to the Pb–Pb data, or to the initial $(^{26}\text{Al}/^{27}\text{Al})_{\text{ss}}$, $(^{53}\text{Mn}/^{55}\text{Mn})_{\text{ss}}$ or $(^{182}\text{Hf}/^{180}\text{Hf})_{\text{ss}}$ -values. Uncertainties are reported as standard errors.

5 Quantifying dispersion

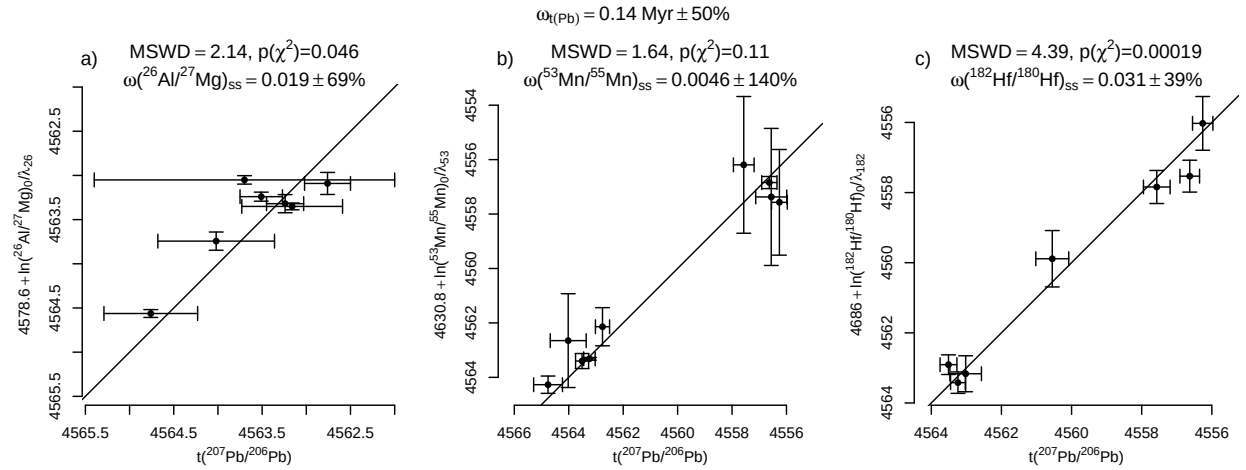


Figure 3: Similar to Figure 2, but using the adjusted Pb–Pb dates of Table 1.

The observed overdispersion can be mathematically captured by random effects models that attribute the excess dispersion to either the x- or the y-variables of Figures 2 and 3. The problem can be formalised by introducing the following definitions:

$t_x(m, i)$: the measured Pb–Pb date of the i^{th} aliquot;

$t_y(m, i)$: the measured Al–Mg, Mn–Cr or W–Hf date of the i^{th} aliquot;

$\delta_x(i)$ and $\delta_y(i)$: the offset between the true x- and y-value caused by dispersion of the true Pb–Pb age, or by the heterogeneity of $(^{26}\text{Al}/^{27}\text{Al})_{SS}$, $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$ or $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$, respectively;

$x(i)(+\delta_x(i))$: the true Pb–Pb age of the i^{th} aliquot;

$y(i)(+\delta_y(i))$: the true Al–Mg, Mn–Cr or Hf–W age of the i^{th} aliquot;

$\epsilon_x(i), \epsilon_y(i)$: residuals, caused by analytical uncertainty.

Using these definitions, two random effects models can be formulated:

$$\text{Model 3x: } \begin{cases} t_x(m, i) = x(i) + \delta_x(i) + \epsilon_x(i) \\ t_y(m, i) = x(i) + \epsilon_y(i) \end{cases}, \text{ Model 3y: } \begin{cases} t_x(m, i) = y(i) + \epsilon_x(i) \\ t_y(m, i) = y(i) + \delta_y(i) + \epsilon_y(i) \end{cases} \quad (3)$$

To keep the problem tractable, it is necessary to make the simplifying assumption that $\delta_x(i)$, $\delta_y(i)$, $\epsilon_x(i)$ and $\epsilon_y(i)$ were drawn from normal distributions with zero mean. For $\epsilon_x(i)$ and $\epsilon_y(i)$, this assumption is reasonable. For $\delta_x(i)$ and $\delta_y(i)$, that is not necessarily so. For example, it is possible that there were multiple reservoirs with distinctly different $(^{26}\text{Al}/^{27}\text{Al})_{SS}$ -values. However, Figure 3a) only contains 7 double-dates, which is too small to identify such multi-modality. Therefore, the simple normal assumption is necessary. The model can easily be modified to capture more realistic scenarios should a sufficiently large database of multi-dated meteorites become available in the future.

For model 3x, we will assume that the $\delta_x(i)$ -values were drawn from a normal distribution with zero mean and standard deviation $\omega_{t(Pb)}$ (units: Myr). For model 3y, we will assume that the $\delta_y(i)$ -values were drawn from a normal distribution with zero mean and standard deviation $\omega_y \lambda_y$ (units: none), which captures the dispersion of the $(^{26}\text{Al}/^{27}\text{Al})_{SS}$, $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$ or $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$ -values, depending on the dataset used to solve the problem. Note that model 3x can be solved using the combined dataset of Al–Mg, Mn–Cr and Hf–W measurements.

The analysis shows that the overdispersion of the adjusted meteorite dataset can either be explained by:

1. 0.5–7.5% heterogeneity of the $(^{26}\text{Al}/^{27}\text{Al})_{SS}$ -ratio (at 2σ , i.e. $\exp[\ln(0.019) \pm 2 \times 0.69]$; Figure 3a).
2. 2–7.5% heterogeneity of the $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$ -ratio;
3. 1.4–6.8% heterogeneity of the $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$ -ratio;

or by

4. 50–380 ka excess dispersion in the Pb–Pb isochron data, caused by partial resetting during transient heating events (Desch et al., 2023a) or by possible over-zealous outlier rejection, which unduly promotes precision over accuracy (Vermeesch, 2026).

Note that, if the original Pb–Pb data are used, then the dispersion increases to 40–130% for $(^{26}\text{Al}/^{27}\text{Al})_{SS}$, 5–22% for $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$, 10–10% for $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$ and 450–900 kyr for t_{Pb} .

6 Conclusion

This paper presented two equivalent approaches to statistical chronometry, using an existing compilation of multi-dated meteorites originally presented by Desch et al. (2023a). The first approach tracks events forward in time from t_{SS} , which is treated as an unknown parameter, and is estimated assuming an assumed $(^{26}\text{Al}/^{27}\text{Al})_{SS}$ -value of 5.23×10^{-5} along with $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$, $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$ and (optionally) λ_{53} . Using the correct number of degrees of freedom, this approach reveals statistically significant dispersion between the extinct and extant chronometers. The second approach uses bivariate linear regression of double-dated meteorites to assess the concordance of the meteorite database. It does not treat t_{SS} , $(^{53}\text{Mn}/^{55}\text{Mn})_{SS}$ and $(^{182}\text{Hf}/^{180}\text{Hf})_{SS}$ as explicit unknowns, but instead converts the $\ln(^{26}\text{Al}/^{27}\text{Al})_0$, $\ln(^{53}\text{Mn}/^{55}\text{Mn})_0$ and $\ln(^{182}\text{Hf}/^{180}\text{Hf})_0$ logratios to absolute ages in ‘Pb–Pb time’ using a linear transformation (Equation 2). Applying the second approach to the Desch et al. (2023a) leads to the same conclusion as the first approach, revealing statistically significant dispersion of the extinct and the extant chronometers.

The regression approach marks a step forward in the development of a statistical chronometry for meteorites, by going from a simple test for statistical homogeneity, to the quantification of heterogeneity. This is important because, to paraphrase statistician George Box “all effects are significant (in some decimal place)”. Given a sufficiently large or sufficiently precise database of multi-dated meteorites, even the smallest degree of heterogeneity eventually becomes detectable. The chi-square test cannot determine whether this heterogeneity has an astrophysical or analytical cause. Likely, a combination of both factors is at play. In order to disentangle the relative contributions of these two possibilities, improved analytical and statistical approaches are needed.

Analytical improvements include the addition of $^{238}\text{U}/^{235}\text{U}$ measurements in the same mineral phase as the Pb-isotope measurements. Although this is challenging to do in small or U-poor samples, it is highly beneficial as it removes the need for the nominal 0.3‰ correction applied by Desch et al. (2023a,b) to many of the meteorites in Table 1. Statistical improvements include a more rigorous selection of aliquots to include in Pb–Pb isochrons. For example, Desch et al. (2023a) present a detailed discussion of Schiller et al. (2015)’s Pb–Pb dataset for NWA 1670 and show that there are at least three statistically equivalent ways to form an isochron through these data, with significantly different ages. To solve this problem, it would be useful to abandon the binary notion that aliquots fall either ‘on’ or ‘off’ the concordia line. A statistically more rigorous approach would be to assign a finite probability to both categories for each aliquot, similar to the random effects model of Section 5. Such a ‘leftmost isochron’ algorithm was recently implemented in **IsoplotR** (Vermeesch, 2018) and will be formally discussed in a forthcoming publication.

The future addition of more multi-dated meteorites to Desch et al. (2023b)’s compilation will make it possible to replace the simple random effects model of Section 5 with more realistic alternatives. For example, given a sufficient number of multi-dated NN and CC chondrites, it would be possible to quantify the within-group and between-group variability of these two meteorite classes. Thus, the nascent field of statistical chronometry has the potential to test physics-based predictions of solar-system formation and provide new insights into its temporal evolution.

Acknowledgments

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